

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2013-14

Paper No. Ancillary Mathematics II

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. (Compulsory — 2 marks each.)

- (a) If $f(x) = |x|$, find $f(-1)$.
- (b) Prove that $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist.
- (c) Examine the continuity of the function:

$$f(x) = \begin{cases} 1 + x, & x \leq 2 \\ 5 - x, & x > 2 \end{cases}$$

at $x = 2$.

- (d) Find the value of k if the function:

$$f(x) = \begin{cases} x - k, & x > 2 \\ kx^2, & x \leq 2 \end{cases}$$

is continuous at $x = 2$.

- (e) Write the necessary condition for a function $f(x)$ to have a local maximum at $x = a$.
- (f) State the order and the degree of the following differential equation:

$$x \frac{d^2 y}{dx^2} + y \left(\frac{dy}{dx} \right)^2 + x = 0$$

- (g) Evaluate the definite integral:

$$\int_0^2 (t^2 + 1) dt$$

- (h) Show that the function $f(x) = |x|$ is not differentiable at $x = 0$.

- (i) Differentiate the following function with respect to x :

$$y = (ax)^n + b^m$$

- (j) Find the points on the curve $y = x^3 - 3x^2 + 3x$, where the gradient (slope) is zero.

Question 2.

(a) Evaluate the limit:

$$\lim_{x \rightarrow -1} \frac{x^2 - 1}{x^2 + 3x + 2}$$

[6]

(b) A function $f(x)$ is defined as:

$$f(x) = \begin{cases} 1, & \text{when } x \neq 0 \\ 0, & \text{when } x = 0 \end{cases}$$

Does the limit of $f(x)$ as $x \rightarrow 0$ exist? Justify your answer.

[6]

Question 3.

(a) Show that the function $f(x)$ defined as:

$$f(x) = \begin{cases} \frac{x^2-1}{x-1}, & x \neq 1 \\ 2, & x = 1 \end{cases}$$

is continuous at $x = 1$.

[6]

(b) Let:

$$f(x) = \begin{cases} x^2, & \text{if } x \leq 1 \\ 2, & \text{if } x > 1 \end{cases}$$

Show that $f(x)$ is discontinuous at $x = 1$.

[6]

Question 4.

(a) Find the derivative with respect to x of the function:

$$y = \frac{x^3 + x^2}{x^2 + 1}$$

[6]

(b) If $y = ax + \frac{b}{x}$, then prove that:

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = 0$$

[6]

Question 5. Find the following integrals:

[12]

(a) $\int (x+1)(2x+3) dx$

[4]

(b) $\int \frac{dx}{x^2 - a^2}$

[4]

(c) $\int \sqrt{2x+5} \, dx$ [4]

Question 6. Evaluate the following definite integrals: [12]

(a) $\int_1^2 \left(x + 3x^2 - \frac{6}{x} \right) dx$ [6]

(b) $\int_0^{\pi/2} (\sin x - \cos x) dx$ [6]

Question 7.

(a) Solve the following ordinary differential equations: [8]

(i) $\frac{dy}{dx} = 1 + y$

(ii) $x \frac{dy}{dx} = x + y$

(b) Solve the ordinary differential equation:

$$(x^2 + 1) \frac{dy}{dx} = 1 + y^2$$

[4]